

1 Chapter 1

Nick's Question 1.1 Proof of lemma 1.8.4

The proof is very nice.

2 Chapter 2

Nick's Question 2.1 In the proof of Theorem 2.4.3 on the bottom of page 34 there are some claims made that I am unable to verify why they are true. First, we are considering

$$X_t = X_0 + \int_0^t u(s)dB_s$$

and here $u(s)$ only satisfies

$$\mathbb{P}\left(\int_0^t u(s)^2 ds < \infty\right) = 1$$

so, as in the Kuo text, this stochastic integral exists and a limit in probability of a sequence of stochastic integrals $\int_0^t u_n(s)dB_s$ where the u_n are simple and are in L^2_{ad} and hence $\int_0^t u_n(s)dB_s$ is a martingale and we know the first and second moment. But for the limit $\int_0^t u(s)dB_s$ we know nothing about the moments. But in this text, it is claimed that

$$\mathbb{E}\left[\int_{t_i}^{t_{i+1}} u^2(s)ds - \left(\int_{t_i}^{t_{i+1}} u(s)dB_s\right)^2 \middle| \mathcal{F}_i\right] = 0$$

and I have never seen this claimed in any other text.

Note that u is bounded (we are assuming localization has been done). Hence, all moments exists.

Nick's Question 2.2 In the proof of theorem 2.5.1, why is it true that

$$E\left(\int_0^t |u'_n(B_s) - 1_{(x,\infty)}(B_s)|^2 ds\right) \leq \int_0^t \mathbb{P}\left(|B_s - x| \leq \frac{2}{n}\right) ds$$

Please explore the mollifier f and f_n . Plot these u'_n and see how could it differ from $1_{(x,\infty)}$.

Nick's Question 2.3 What is the motivation on page 41 behind the following definition of the backward ito integral

$$\int_0^T u(s)\hat{d}B_s = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-2} \left(\frac{n}{T} \int_{(j+1)T/n}^{(j+2)T/n} u(s)ds\right) (B_{(j+1)T/n} - B_{jT/n})$$

The integral in the parenthesis is the average of u_s on $((j+1)T/n, (j+2)T/n)$.

Let's skip this part. I know nothing on the backward SDE and not particular interested either. It has applications in finance. If you don't do finance, I don't see the point here.

Nick's Question 2.4 (Problem 2.7)

This problem follows easily from Levy's Characterization of Brownian Motion. But I can't figure out how to do this problem with the hint given.

This is a nice problem. The hints is to ask you to find the distribution of $X_t - X_s$ through the characteristic function. Once apply Ito's formula, take expectation and remove the martingale part, you should arrive at an integral equation. Take derivatives, you arrive at a differential equation. The solution for $a_s(t) := \mathbb{E}(e^{i\lambda(X_t - X_s)})$ is

$$a_s(t) = e^{-\frac{\lambda^2}{2}(t-s)}.$$

Nick's Question 2.5 (Problem 2.15)

First we consider the 2 - d Ito process

$$\begin{pmatrix} M_s \\ N_s \end{pmatrix} = \begin{pmatrix} E(X^p) \\ E(Y^{q'}) \end{pmatrix} + \int_0^t \begin{pmatrix} \varphi_s & 0 \\ 0 & \psi_s \end{pmatrix} \begin{pmatrix} d\hat{B}_s \\ dB_s \end{pmatrix}$$

and then following the hint we apply Ito's formula to $f(x, y) = x^{1/p}y^{1/q'}$. We get the following derivatives:

- $f_x = \frac{1}{p}x^{\frac{1}{p}-1}y^{\frac{1}{q'}}$
- $f_{xx} = \frac{1}{p}\left(\frac{1}{p}-1\right)x^{\frac{1}{p}-2}y^{\frac{1}{q'}}$
- $f_y = \frac{1}{q'}x^{\frac{1}{p}}y^{\frac{1}{q'}-1}$
- $f_{yy} = \frac{1}{q'}\left(\frac{1}{q'}-1\right)x^{\frac{1}{p}}y^{\frac{1}{q'}-2}$

Now putting this into Ito's Formula gives us

$$\begin{aligned} M_t^{1/p} N_t^{1/q'} &= \|X\|_p \|Y\|_{q'} + \frac{1}{p} \int_0^t M_s^{\frac{1}{p}-1} N_s^{\frac{1}{q'}} \varphi_s d\hat{B}_s + \frac{1}{q'} \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-1} \psi_s dB_s \\ &\quad + \frac{1}{2} \frac{1}{p} \left(\frac{1}{p}-1\right) \int_0^t M_s^{\frac{1}{p}-2} N_s^{\frac{1}{q'}} \varphi_s^2 ds + \frac{1}{2} \frac{1}{q'} \left(\frac{1}{q'}-1\right) \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-2} \psi_s^2 ds \end{aligned}$$

I am not really sure where to continue from here. I see that if we let $t \rightarrow \infty$ then $M_t^{1/p} N_t^{1/q'} \rightarrow XY$ so if we took the expected value of both sides then we would partially have what we are looking for since $E(XY)$ is on the left and $\|X\|_p \|Y\|_{q'}$ is on the right. But I am not sure about the other terms. Im guessing the Ito integrals in the right have 0 expectation but I am not sure if that is in fact true since I dont see how it is easily seen that the integrands are in $L^2(\mathcal{P})$.

What I did next was see what $\hat{B}_s$ was equal too and for this we define another Ito Process

$$\begin{pmatrix} B_s \\ \tilde{B}_s \end{pmatrix} = \int_0^t \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} dB_s \\ d\tilde{B}_s \end{pmatrix}$$

and applied Itos formula to $f(t, x, y) = e^{-t}x + \sqrt{1 - e^{-2t}}y$ which gives us

$$\hat{B}_s = e^{-t}B_s + \sqrt{1 - e^{-2t}}\tilde{B}_s = \int_0^t \frac{e^{-2s}}{\sqrt{1 - e^{-2s}}}\tilde{B}_s - e^{-s}B_s ds + \int_0^t e^{-s}dB_s + \int_0^t \sqrt{1 - e^{-2s}}d\tilde{B}_s$$

However I am not sure if this step is necessary.

The followings are some of my thoughts: First of all, we need the absolute values:

$$\begin{pmatrix} M_s \\ N_s \end{pmatrix} = \begin{pmatrix} \mathbb{E}(|X|^p) \\ \mathbb{E}(|Y|^{q'}) \end{pmatrix} + \int_0^t \begin{pmatrix} \varphi_s & 0 \\ 0 & \psi_s \end{pmatrix} \begin{pmatrix} d\hat{B}_s \\ dB_s \end{pmatrix}.$$

Second, you forget

$$f_{xy} = \frac{1}{p} \frac{1}{q'} x^{-1/(p')} y^{-1/q}.$$

Hence, by Ito's formula, we will have

$$\begin{aligned} M_t^{1/p} N_t^{1/q'} &= \|X\|_p \|Y\|_{q'} + \frac{1}{p} \int_0^t M_s^{\frac{1}{p}-1} N_s^{\frac{1}{q'}} \varphi_s d\hat{B}_s + \frac{1}{q'} \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-1} \psi_s dB_s \\ &\quad + \frac{1}{2} \frac{1}{p} \left(\frac{1}{p} - 1 \right) \int_0^t M_s^{\frac{1}{p}-2} N_s^{\frac{1}{q'}} \varphi_s^2 ds + \frac{1}{2} \frac{1}{q'} \left(\frac{1}{q'} - 1 \right) \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-2} \psi_s^2 ds \\ &\quad + \frac{1}{pq'} \int_0^t M_s^{-\frac{1}{p'}} N_s^{-\frac{1}{q}} \psi_s \phi_s d\langle B, \hat{B} \rangle_s \\ &=: \|X\|_p \|Y\|_{q'} + \sum_{i=1}^5 I_i. \end{aligned}$$

Upon taking expectation on both sides, we see that

$$LHS = \mathbb{E}(|X_t Y_t|) \geq \mathbb{E}(X_t Y_t).$$

We only need to prove that

$$\mathbb{E} \left(\sum_{i=1}^5 I_i \right) \leq 0,$$

which is clear for $i = 1, 2$, in which case we have the expectation equal to zero. Now for the rest three parts, we see that

$$I_3 + I_4 + I_5 = -\frac{1}{2pq'} M_s^{\frac{1}{p}-2} N_s^{\frac{1}{q'}-2} \left(N_s^2 \varphi_s^2 \frac{q'}{p'} - 2M_s N_s \phi_s \psi_s e^{-s} + M_s \psi_s^2 \frac{p}{q} \right),$$

where we have used the fact that $d\langle B, \hat{B} \rangle_s = e^{-s} ds$. Hence, the condition become the condition that the quadratic form has no solution:

$$\Delta = 4e^{-2s} - 4 \frac{q'p}{p'q} = 4 \left(e^{-2s} - \frac{p-1}{q-1} \right) \leq 0$$

which is equivalent to

$$q \leq e^{2s}(p-1) + 1.$$

Hence, the largest q that makes the inequality work is $q = e^{2s}(p-1) + 1$.

3 Chapter 3

Nick's Question 3.1 (Smooth Cylindrical Random Variable) Following chapter 3 of the Kuo text, denote C the Wiener space of continuous functions from $[0,1]$ to $\mathbb{R}$, $\mathcal{R}$ the collection cylindrical sets

$$A = \{\omega \in C : (\omega(t_1), \dots, \omega(t_n)) \in U \mid t_i \in [0, 1], n \in \mathbb{N}, U \in \mathcal{B}(\mathbb{R}^n)\}$$

and μ the associated Wiener measure. Consider the coordinate processes $B(t, \omega) = \omega(t)$ which are Brownian motions under $(C, \mathcal{B}(\mathcal{R}), \mu)$. There is a theorem that can be found [here](#).

Theorem : The random variables

$$\{f(B(t_1, \cdot), \dots, B(t_n, \cdot)) \mid n \in \mathbb{N}, f \in \mathcal{S}(\mathbb{R}^n)\}$$

are dense in $L^2(C, \mu)$ where $\mathcal{S}(\mathbb{R}^n)$ are the Schwartz functions.

How does this relate to $\mathcal{S}$ defined on page 51 of the intro to Malliavin textbook? How are the cylindrical random variables defined in this case? Why is $\mathcal{S}$ dense in $L^2(\Omega)$? Even in Diffusions Markov Processes and Martingales - Roger and Williams, they define the cylindrical sigma field in the same way Kuo does- through the use of coordinate processes.

Nick's Question 3.2 Why is $(D_t F)_{t \geq 0}$ a stochastic processes in $L^2(\Omega; H)$ where $H = L^2(\mathbb{R}^+)$? $D_t F$ is defined as

$$D_t F = \sum_{i=1}^n \frac{\partial f}{\partial x}(B(h_1), \dots, B(h_n)) h_i(t)$$

where $h_i \in H$ and $B(h_i) = \int_0^\infty h_i dB_s$ is the standard Wiener Integral. So isnt $D_t F$ real valued since we are evaluating the $h_i(\cdot)$ at t ? Then shouldn't the $(D_t F)_{t \geq 0}$ be a $L^2(\Omega; \mathbb{R})$ valued stochastic process?

Nick's Question 3.3 (Proposition 3.2.4)

1. Why does $D_h(\delta(u)) = \langle u, h \rangle + \delta(D_h u)$? (I have answered this below)
2. Also, at the bottom of page 52, why is it that (I have answered this below)

$$E \left(\sum_{i,j=1}^{\infty} D_{e_i} \langle u, e_j \rangle D_{e_j} \langle v, e_i \rangle \right) = E \left(\int_0^\infty \int_0^\infty D_s u_t D_t v_s ds dt \right)$$

3. In the proof of property 3.3 why do we have

$$E \left(\sum_{i=1}^{\infty} \langle v, e_i \rangle \left(\langle u, e_i \rangle + \delta(D_{e_i}(u)) \right) \right) = E(\langle u, v \rangle) + E \left(\sum_{i,j=1}^{\infty} D_{e_i} \langle e, e_j \rangle D_{e_j} \langle v, e_i \rangle \right)$$

Proof of $D_h(\delta(u)) = \langle u, h \rangle + \delta(D_h u)$.

Since $u = \sum F_j h_j$ we just apply the definition of δ to get

$$D_h(\delta(u)) = D_h \left(\sum_{j=1}^n F_j B(h_j) - \sum_{j=1}^n \langle D F_j, h_j \rangle \right)$$

$$\begin{aligned}
&= \sum_{j=1}^n D_h(F_j B(h_j)) - \sum_{j=1}^n D_h \langle DF_j, h_j \rangle \\
&= \sum_{j=1}^n F_j \langle h, h_j \rangle + \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n D_h \langle DF_j, h_j \rangle
\end{aligned} \tag{3.1} \quad \boxed{\text{P:3.2.4-1}}$$

where the third inequality follows from the product rule and $D_h(B(h_j)) = \langle h, h_j \rangle$. Now we focus on the last term from the last line above (3.1).

$$\sum_{j=1}^n D_h \langle DF_j, h_j \rangle = \sum_{j=1}^n \langle D_t \langle DF_j, h_j \rangle, h(t) \rangle \tag{3.2} \quad \boxed{\text{P:3.2.4-6}}$$

Before we continue, we simplify $D_t \langle DF_j, h_j \rangle$. We may assume that $F_j = f^j(B(h_1^j(t), \dots, B(h_n^j(t))))$ where $f^j \in C_p^\infty(\mathbb{R}^n)$ and $h_i^j \in H$.

$$\begin{aligned}
D_t \langle DF_j, h_j \rangle &= D_t \left\langle \sum_{i=1}^n \frac{\partial f^j}{\partial x_i} h_i^j(s), h_j(s) \right\rangle \\
&= \sum_{i=1}^n \langle h_i^j(s), h_j(s) \rangle D_t \left(\frac{\partial f^j}{\partial x_i} \right)
\end{aligned} \tag{3.3} \quad \boxed{\text{P:3.2.4-2}}$$

We now plug (3.3) into (3.2) above to get

$$\begin{aligned}
\sum_{j=1}^n \langle D_t \langle DF_j, h_j \rangle, h(t) \rangle &= \sum_{j=1}^n \sum_{i=1}^n \langle h_i^j(s), h_j(s) \rangle \langle D_t \left(\frac{\partial f^j}{\partial x_i} \right), h(t) \rangle \\
&= \sum_{j=1}^n \sum_{i=1}^n \left\langle h_i^j(s) \langle D_t \left(\frac{\partial f^j}{\partial x_i} \right), h(t) \rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \left\langle D_t \left(\sum_{i=1}^n h_i^j(s) \frac{\partial f^j}{\partial x_i} \right), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \langle D_t(DF_j), h(t) \rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle D_h(DF_j), h_j(s) \right\rangle
\end{aligned}$$

and if I haven't made a mistake, then what we have shown so far is that

$$\begin{aligned}
D_h(\delta(u)) &= \sum_{j=1}^n F_j \langle h, h_j \rangle + \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n D_h \langle DF_j, h_j \rangle \\
&= \sum_{j=1}^n F_j \langle h, h_j \rangle + \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n \left\langle D_h(DF_j), h_j \right\rangle
\end{aligned} \tag{3.4} \quad \boxed{\text{P:3.2.4-3}}$$

Also, the above is exactly line two of the first centered equations on page 53.

Now we need to show that the last two terms add up to $\delta(D_h(u))$. This is not defined in the text but I think that we need to define $D_h u$ as the following

$$D_h u = D_h \sum_{j=1}^n F_j h_j(t) = \sum_{j=1}^n h_j(t) D_h(F_j) \tag{3.5} \quad \boxed{\text{P:3.2.4-4}}$$

Thus now $D_h(u)$ (3.5) is of the form $\sum F_j h_j$ and so we may apply the definition of δ to it to get the following

$$\delta(D_h u) = \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n \langle D_s(D_h F_j), h_j(s) \rangle \quad (3.6) \quad \boxed{\text{P:3.2.4-5}}$$

Now we focus in the last term of the right hand side of (3.6). From (3.4) we note that we want to show that

$$\sum_{j=1}^n \langle D_s(D_h F_j), h_j(s) \rangle = \sum_{j=1}^n \left\langle D_h(D F_j), h_j \right\rangle \quad (3.7) \quad \boxed{\text{P:3.2.4-7}}$$

Here are the steps,

$$\begin{aligned} \sum_{j=1}^n \langle D_s(D_h F_j), h_j(s) \rangle &= \sum_{j=1}^n \left\langle D_s \left(\left\langle \sum_{i=1}^n \frac{\partial f^j}{\partial x_i} h_i^j(t), h(t) \right\rangle \right), h_j(s) \right\rangle \\ &= \sum_{j=1}^n \left\langle D_s \left(\sum_{i=1}^n \frac{\partial f^j}{\partial x_i} \langle h_i^j(t), h(t) \rangle \right), h_j(s) \right\rangle \\ &= \sum_{j=1}^n \left\langle \sum_{i=1}^n D_s \left(\frac{\partial f^j}{\partial x_i} \right) \langle h_i^j(t), h(t) \rangle, h_j(s) \right\rangle \\ &= \sum_{j=1}^n \left\langle \left\langle \sum_{i=1}^n D_s \left(\frac{\partial f^j}{\partial x_i} \right) h_i^j(t), h(t) \right\rangle, h_j(s) \right\rangle \\ &= \sum_{j=1}^n \left\langle \left\langle \sum_{i=1}^n D_s \left(\frac{\partial f^j}{\partial x_i} h_i^j(t) \right), h(t) \right\rangle, h_j(s) \right\rangle \\ &= \sum_{j=1}^n \left\langle \langle D_s(D_t F_j), h(t) \rangle, h_j(s) \right\rangle \\ &= \sum_{j=1}^n \left\langle D_h(D F_j), h_j \right\rangle \end{aligned}$$

Hence (3.7) is shown and we are done.

Proof of bottom of page 52

$$\begin{aligned} \sum_{i,j=1}^{\infty} D_{e_i} \langle u, e_j \rangle D_{e_j} \langle v, e_i \rangle &= \sum_{i,j=1}^{\infty} \left\langle D_s \langle u, e_j \rangle, e_i(s) \right\rangle \left\langle D_t \langle v, e_i \rangle, e_j(t) \right\rangle \\ &= \sum_{i,j=1}^{\infty} \left\langle \langle D_s(u_t), e_j(t) \rangle, e_i(s) \right\rangle \left\langle \langle D_t(v_s), e_i(s) \rangle, e_j(t) \right\rangle \\ &= \sum_{i=1}^{\infty} \left\langle \sum_{j=1}^{\infty} \left(\langle D_s(u_t), e_j(t) \rangle \left\langle \langle D_t(v_s), e_i(s) \rangle, e_j(t) \right\rangle \right), e_i(s) \right\rangle \\ &= \sum_{i=1}^{\infty} \left\langle \left\langle D_s(u_t), \langle D_t(v_s), e_i(s) \rangle \right\rangle, e_i(s) \right\rangle \\ &= \sum_{i=1}^{\infty} \left\langle \left\langle D_s(u_t), \langle D_t(v_q), e_i(q) \rangle \right\rangle, e_i(s) \right\rangle \end{aligned}$$

$$\begin{aligned}
&= \sum_{i=1}^{\infty} \int_0^{\infty} \left(\int_0^{\infty} D_s(u_t) \left(\int_0^{\infty} D_t(v_q) e_i(q) dq \right) dt \right) e_i(s) ds \\
&= \int_0^{\infty} \left(\int_0^{\infty} D_s(u_t) \left(\sum_{i=1}^{\infty} \int_0^{\infty} D_t(v_q) e_i(q) dq e_i(s) \right) dt \right) ds \\
&= \int_0^{\infty} \int_0^{\infty} D_s(u_t) D_t(v_s) dt ds
\end{aligned}$$

Where above we use the identity

$$\langle f, g \rangle = \sum_{i=1}^{\infty} \langle f, e_i \rangle \langle g, e_i \rangle.$$

Nick's Question 3.4 (Why is δ linear?) for $u = \sum_{j=1}^n F_j h_j$ we have

On page 55 it says that δ is linear. First,

$$\delta(u) = \sum_{j=1}^n F_j B(h_j) - \sum_{j=1}^n \langle DF_j, h_j \rangle_H$$

Now if we have a $v = \sum_{j=1}^n G_j h'_j$, how does $\delta(u + v) = \delta(u) + \delta(v)$. Also how could we write $\sum_{j=1}^n F_j h_j + \sum_{j=1}^n G_j h'_j = \sum_{j=1}^n K_j k_j$ where $K_j \in \mathcal{S}$ and $k_j \in H$

Do we do this through the relationship $E(\langle DF, u \rangle) = E(\delta(u)F)$ in the sense that

$$\begin{aligned}
E(\delta(u + v)F) &= E(\langle DF, u + v \rangle) \\
&= E(\langle DF, u \rangle) + E(\langle DF, v \rangle) \\
&= E(\delta(u)F) + E(\delta(v)F) \\
&= E((\delta(u) + \delta(v))F)
\end{aligned}$$

and if this is true for every F then I believe we can say $\delta(u + v) = \delta(u) + \delta(v)$?